

PROGRAM ASSESSMENT REPORT

v2024.1

Tracking data

CIM code	BS-CS-2, CS-CIS-2
Program name	BS in Computer Science, BS in Computer Information Systems
year	Fall 2025
Program assessment coordinator	Matthew Bauer
Assessment plan year	2024-2025
No assessment this year per plan	
LO(s) assessed	A subset of the ABET Program Outcomes that map to CS425. Both BS in CS and BS in CIS have these outcomes: 1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions. 2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline 5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline. 6. Apply computer science theory and software development fundamentals to produce computing-based solutions.
Presented at program authority meeting date: <i>Please attach minutes</i>	CS UG Studies Committee, Fall 2025

Methodology

For each LO and artifact assessed, please describe methodology used, and attach (as supplemental files during report submission) or insert here the rubric used. At minimum this should contain:

- a description of how the artifact is concordant with the LO
- a rubric describing the method used to place individual student in at least 3 assessment categories; {**D** Does not meet expectations, **M** Meets expectations **M**, **P** Proficient}

All LOs are assessed on a 5-point scale (Good-5 points, Average-3 points, Below Average-1 point) using a random 20% of BS in CS/CIS students enrolled in CS425 in Fall 2024 and Spring 2025. The target thresholds are 70% of the students tested should achieve a score of 3 or better (out of possible 5 points) OR 7 or better (out of possible 10 points). The artifacts are student homework, project deliverables, and exam questions that map to each outcome/rubric.

Outcomes/Rubrics

1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.

Good (5 points each)

Student consistently demonstrated strong knowledge of both relational algebra and SQL syntax.

Average (3 points each)

Student demonstrated knowledge of relational algebra and SQL syntax, but not consistently.

Below Average (1 pt each)

Student demonstrated little to no knowledge of both relational algebra and SQL syntax.

2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.

Good (5 points each)

Student consistently demonstrated ability to apply database design and entity relationship design to a set of requirements.

Average (3 points each)

Student demonstrated ability to apply database design and entity relationship design to a set of requirements, but not consistently.

Below Average (1 pt each)

Student demonstrated little to no ability to apply database design and entity relationship design to a set of requirements.

5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.

Good (5 points each)

All deliverables on time and satisfactory

Average (3 points each)

More than half (but not all) the deliverables on time and satisfactory

Below Average (1 pt each)

Less than half the deliverables on time and satisfactory

6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Good (5 points each)

Students can apply more than one design and development principle in the construction of software systems.

Average (3 points each)

Students can apply one design and development principle in the construction of software systems.

Below Average (1 pt each)

Students recall at least one design and development principle but cannot apply it in the construction of software systems.

Results

Artifact samples

Attach artifact samples. For data sets with ≤ 10 students, provide 1 in each category (unless the category is empty, if so state that). For programs with ≤ 20 , provide 2. For other programs provide 3. Please redact any personally identifiable information, PII (which included names and A#s).

[F24 - CS425 Student Artifacts](#)

[S25 - CS425 Student Artifacts](#)

Data table

For each LO and provide a table pasted in from the [provided template](#) placing each student into the assessment categories (minimally {D, M, P}). Please redact any personally identifiable information, PII (including names or A#s) from this report. Provide a summary graph from the template presenting this distribution. You should also present additional analysis as deemed appropriate by the results seen, and the needs of the program.

[CS425 ABET F24 S25.xlsx](#)

The target thresholds are 70% of the students tested should achieve a score of 3 or better (out of possible 5 points) OR 7 or better (out of possible 10 points).

Outcome/Rubric	F24 Passing Threshold	S25 Passing Threshold
1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.		
Student consistently demonstrated strong knowledge of both relational algebra and SQL syntax.	100%	60%
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.		
Student consistently demonstrated ability to apply database design and entity relationship design to a set of requirements.	70%	80%
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		
All deliverables on time and satisfactory	100%	90%
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.		
Students can apply more than one design and development principle in the construction of software systems.	100%	100%

Description of findings

Summarize the discussion that occurred within the program authority when this was presented. Provide a short narrative interpreting the findings. Are there any areas of concern?

Seven of the eight rubrics/student populations passed the 70% threshold. One student population did not achieve the threshold for the "Student consistently demonstrated strong knowledge of both relational algebra and SQL syntax." rubric.

Continuous Improvement, CI, plans

The purpose of assessment is to improve the quality of our educational programs. The most important part of the assessment cycle is how we use the data to improve our programs. Examine the data provided above describing student achievement, and if areas for possible improvement emerge, describe plans to achieve this improvement.

The one rubric below threshold was assessed twice, once on "S25 HW1" near the start of the term, and once on "S25 Project Deliverable #2" near the middle of the term. Since we do see equal or better

performance for 9 of the 10 students near the middle of the term, the CS UG Studies Committee determined that no action was necessary.

Prior CI plan impact

Also, please refer to the previous assessment report for the LOs assessed. Were any CI plans recommended? Were they implemented? Did the plans work (were improvements seen in this cycle)?

☐no previous CI plans available

No issues from the previous assessment cycle that considered these outcomes.

This cycle's CI plan

Describe any CI plans developed from this cycle's data, as well as the strategy developed to implement these plans.

After consideration and presentation to the CS UG Studies Committee, no CI plans are recommended.